HE2001 Microeconomics II

Tutorial 2

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Overview

This tutorial covers:

- revealed preference axioms (WARP, GARP, SARP) and their relationships;
- strict concavity and uniqueness of utility maximisers on budget sets;
- implications of price changes under GARP;
- comparative statics and Slutsky decomposition for a given Marshallian demand;
- perfect complements: indifference curves, Marshallian demand, and Slutsky effects.

1 GARP

Bob makes purchases to maximize an increasing utility function and therefore his choices must satisfy GARP. We observe Bob buy $x = (x_1, x_2)$ when prices are $p = (p_1, p_2)$ where $x_1 > 0$ and $x_2 > 0$ and he spends $m = p \cdot x$.

1.1 WARP but not GARP with three goods

I told you that GARP and WARP are equivalent when there are only two goods. Suppose there are three goods and the consumer buys $x^1 = (1, 0, 0)$, $x^2 = (0, 1, 0)$, and $x^3 = (0, 0, 1)$ at prices p^1, p^2, and p^3. Can you come up with price vectors p^1, p^2, and p^3 so that the consumer's behavior satisfies WARP but not GARP?

Solution

Method 1: Explicit construction and verification by dot products

Take

$$p^1 = (2, 4, 1), \qquad p^2 = (1, 2, 4), \qquad p^3 = (4, 1, 2).$$

Since each x^t is a unit vector, the expenditures are

$$m^1 = p^1 \cdot x^1 = 2, \qquad m^2 = p^2 \cdot x^2 = 2, \qquad m^3 = p^3 \cdot x^3 = 2.$$

Compute revealed preference relations:

$$p^1 \cdot x^3 = 1 \le 2 = m^1 \ \Rightarrow\ x^1 R x^3,$$

$$p^3 \cdot x^2 = 1 \le 2 = m^3 \ \Rightarrow\ x^3 R x^2,$$

$$p^2 \cdot x^1 = 1 \le 2 = m^2 \ \Rightarrow\ x^2 R x^1.$$

Thus $x^1 R x^3 R x^2 R x^1$. Moreover the last step is strict because

$$p^2 \cdot x^1 = 1 < 2 = p^2 \cdot x^2 \ \Rightarrow\ x^2 P x^1,$$

so GARP is violated.

Check WARP (no strict 2-cycle):

$$x^1 R x^3 \text{ but } p^3 \cdot x^1 = 4 > 2 = p^3 \cdot x^3 \ \Rightarrow\ x^3 R x^1 \text{ fails},$$

$$x^3 R x^2 \text{ but } p^2 \cdot x^3 = 4 > 2 = p^2 \cdot x^2 \ \Rightarrow\ x^2 R x^3 \text{ fails},$$

$$x^2 R x^1 \text{ but } p^1 \cdot x^2 = 4 > 2 = p^1 \cdot x^1 \ \Rightarrow\ x^1 R x^2 \text{ fails}.$$

Hence WARP holds but GARP fails.

$$\boxed{p^1 = (2, 4, 1),\ p^2 = (1, 2, 4),\ p^3 = (4, 1, 2) \text{ satisfy WARP but not GARP.}}$$

Method 2: Design inequalities first (a strict 3-cycle without any 2-cycle)

To force $x^1 R x^3$, $x^3 R x^2$, $x^2 R x^1$, it suffices to impose

$$p_3^1 < p_1^1, \qquad p_2^3 < p_3^3, \qquad p_1^2 < p_2^2.$$

To avoid any reverse affordability that could create WARP violations, impose

$$p_1^3 > p_3^3, \qquad p_3^2 > p_2^2, \qquad p_2^1 > p_1^1.$$

Choosing prices consistent with these inequalities yields WARP but not GARP; the numerical choice in Method 1 is one such instance.

$$\boxed{\text{Construct a strict 3-cycle without any strict 2-cycle: WARP holds but GARP fails.}}$$

1.2 Strict concavity implies uniqueness of the maximiser

A utility function $U : \mathbb{R}^L \to \mathbb{R}$ is strictly concave if

$$U\big(\alpha x + (1-\alpha)x'\big) > \alpha U(x) + (1-\alpha)U(x') \tag{1}$$

for all $x' \neq x$ and all α taking values strictly between 0 and 1. Show that if U is strictly concave then each budget set $B(p, m)$ can have at most one consumption bundle which maximizes the utility function. In other words, there is some $x \in B(p, m)$ so that $U(x) > U(x')$ for all $x' \in B(p, m)$ so that $x \neq x'$. *Hint:* suppose there are two bundles x and x' which maximize utility U over all bundles in $B(p, m)$ and show that this contradicts (1).

Solution

Method 1: Midpoint contradiction using strict concavity

Assume for contradiction that there are two distinct maximisers $x \neq x'$ in $B(p, m)$ such that

$$U(x) = U(x') = \max\{U(y) : y \in B(p, m)\}.$$

Since $B(p, m) = \{y \in \mathbb{R}_+^L : p \cdot y \leq m\}$ is convex, the midpoint

$$\bar{x} = \frac{1}{2}x + \frac{1}{2}x'$$

satisfies $\bar{x} \in B(p, m)$ because

$$p \cdot \bar{x} = \frac{1}{2}(p \cdot x) + \frac{1}{2}(p \cdot x') \leq \frac{1}{2}m + \frac{1}{2}m = m.$$

By strict concavity (take $\alpha = \frac{1}{2}$ in (1)) and $x \neq x'$,

$$U(\bar{x}) > \frac{1}{2}U(x) + \frac{1}{2}U(x') = U(x),$$

contradicting maximality of x. Hence there is at most one maximiser.

$$\boxed{\text{If } U \text{ is strictly concave, } B(p, m) \text{ has at most one utility maximiser.}}$$

Method 2: General α-combination contradiction

Assume $x \neq x'$ both maximise U on $B(p,m)$. For any $\alpha \in (0,1)$, define

$$x_\alpha = \alpha x + (1-\alpha)x'.$$

Convexity of $B(p,m)$ implies $x_\alpha \in B(p,m)$. Strict concavity (1) gives

$$U(x_\alpha) > \alpha U(x) + (1-\alpha)U(x') = U(x),$$

contradicting maximality. Thus $x = x'$.

$$\boxed{\text{Strict concavity prevents multiple maximisers on a convex budget set.}}$$

1.3 Strict concavity and revealed-preference cycles

A consumer satisfies the Strong Axiom of Revealed Preferences (SARP) if $x\,R^*\,x'$ and $x'\,R\,x$ then $x = x'$. Suppose that a consumer maximizes a strictly concave utility function (that is, for each t we have $U(x^t) \geq U(x)$ for all $x \in B(p^t,\, p^t \cdot x^t)$) and that

$$x^1 R x^2 R x^3 R x^4 R \cdots R x^T R x^1.$$

Show that $x^1 = x^2 = x^3 = \cdots = x^T$.

Solution

Method 1: Utility chain forces equalities; uniqueness forces equal bundles

For each t, $x^t R x^{t+1}$ means $x^{t+1} \in B(p^t,\, p^t \cdot x^t)$. Since x^t maximises U on that budget,

$$U(x^t) \geq U(x^{t+1}) \quad (t = 1, \ldots, T),$$

where $x^{T+1} = x^1$. Hence

$$U(x^1) \geq U(x^2) \geq \cdots \geq U(x^T) \geq U(x^1),$$

so

$$U(x^1) = U(x^2) = \cdots = U(x^T).$$

Fix t. Because x^{t+1} is feasible in period t and attains the same utility as the chosen maximiser x^t, it must also be a maximiser on the same budget set. By the uniqueness result above, $x^{t+1} = x^t$. Thus all bundles are equal.

$$\boxed{x^1 = x^2 = \cdots = x^T.}$$

Method 2: Strict concavity implies SARP; apply it to the cycle

Under strict concavity, the maximiser on each budget set is unique. If $x^t R x^s$ and $x^s R x^t$, then each is feasible on the other's budget, and optimality implies $U(x^t) \geq U(x^s)$ and $U(x^s) \geq U(x^t)$, hence equality. Uniqueness forces $x^t = x^s$, which is SARP.

In the cycle $x^1 R x^2 R \cdots R x^T R x^1$, for each t we have both $x^1 R x^t$ and $x^t R x^1$. By SARP, $x^t = x^1$ for all t.

$$\boxed{\text{The revealed-preference cycle collapses: } x^1 = x^2 = \cdots = x^T.}$$

1.4 Price increase of good 1 with fixed expenditure implies GARP

When prices are $p = (p_1, p_2)$ Bob purchases $x = (x_1, x_2)$ where $x_1 > 0$. Let $m = p \cdot x$. Prices change to $p' = (p'_1, p_2)$ with $p'_1 > p_1$ (note the price of good 2 is unchanged). Bob buys some bundle x' which satisfies $p' \cdot x' = m$. Show that his behavior satisfies GARP.

Solution

Method 1: Show the old bundle is not affordable under the new prices

Compute:

$$\begin{aligned} p' \cdot x &= p'_1 x_1 + p_2 x_2 \\ &= (p_1 x_1 + p_2 x_2) + (p'_1 - p_1) x_1 \\ &= m + (p'_1 - p_1) x_1 \\ &> m, \end{aligned}$$

since $p'_1 > p_1$ and $x_1 > 0$. Thus $x \notin B(p', m)$, so $x' R x$ is impossible.

But since $p' \cdot x' = m$,

$$p \cdot x' = (p_1 x'_1 + p_2 x'_2) = (p'_1 x'_1 + p_2 x'_2) - (p'_1 - p_1) x'_1 = m - (p'_1 - p_1) x'_1 \leq m,$$

so $x' \in B(p, m)$, hence $x R x'$. Therefore no strict 2-cycle can occur and GARP holds.

$$\boxed{\text{His behaviour satisfies GARP.}}$$

Method 2: Two-observation GARP reduces to "no strict 2-cycle"

With only (p, x) and (p', x'), GARP is violated iff $x R x'$ and $x' P x$, or $x' R x$ and $x P x'$. But $x' R x$ is impossible because $p' \cdot x > m = p' \cdot x'$. Hence no violation is possible.

$$\boxed{\text{GARP holds because a strict 2-cycle cannot form.}}$$

1.5 Compensated expenditure and a GARP implication

As in the previous question suppose that prices change to $p' = (p'_1, p_2)$ with $p'_1 > p_1$ but suppose that expenditure changes to $m' = p' \cdot x$. Suppose the consumer satisfies GARP. Show that his demand x' at the new price p' must satisfy $p \cdot x' \geq p \cdot x$.

Solution

Method 1: Contradiction via a strict revealed-preference cycle

Here $m' = p' \cdot x$ and $p' \cdot x' = m'$, so x is feasible at (p', m'). Hence $x'Rx$.

Assume for contradiction that $p \cdot x' < p \cdot x$. Since x is chosen at prices p with expenditure $p \cdot x$, this implies xPx'. Thus we have xPx' and $x'Rx$, violating GARP. Contradiction.

$$\boxed{p \cdot x' \geq p \cdot x.}$$

Method 2: Once $x'Rx$ holds, GARP forbids xPx'

Since $x'Rx$ holds automatically (because x is affordable at (p', m')), GARP implies not xPx'. But $xPx' \iff p \cdot x' < p \cdot x$. Therefore $p \cdot x' \geq p \cdot x$.

$$\boxed{p \cdot x' \geq p \cdot x.}$$

2 Demand example

Consider the utility function

$$U(x) = \sqrt{x_1} + \sqrt{x_2}.$$

This U generates the demand function

$$x(p_1, p_2, m) = \frac{m}{p_1 + p_2} \left(\frac{p_2}{p_1}, \frac{p_1}{p_2} \right).$$

2.1 Demand at $p = (1, 1)$, $m = 1$

What consumption bundle does the consumer demand when $p = (1, 1)$ and $m = 1$?

Solution

Method 1: Plug into the given demand function

With $p_1 = p_2 = 1$ and $m = 1$,

$$x(1, 1, 1) = \frac{1}{2}(1, 1) = \left(\frac{1}{2}, \frac{1}{2} \right).$$

$$\boxed{x(1,1,1) = \left(\tfrac{1}{2}, \tfrac{1}{2}\right).}$$

Method 2: Lagrangian derivation

Maximise $\sqrt{x_1} + \sqrt{x_2}$ subject to $p_1 x_1 + p_2 x_2 = m$, $x_1, x_2 \geq 0$. Lagrangian:

$$\mathcal{L} = \sqrt{x_1} + \sqrt{x_2} + \lambda(m - p_1 x_1 - p_2 x_2).$$

FOCs (interior):

$$\frac{1}{2\sqrt{x_1}} = \lambda p_1, \qquad \frac{1}{2\sqrt{x_2}} = \lambda p_2 \;\Rightarrow\; \sqrt{x_2} = \frac{p_1}{p_2}\sqrt{x_1}.$$

So $x_2 = (p_1^2/p_2^2)x_1$. Substitute into the budget and solve to obtain the stated demand, hence $(1/2, 1/2)$ at $(1, 1, 1)$.

$$\boxed{x(1,1,1) = \left(\tfrac{1}{2}, \tfrac{1}{2}\right).}$$

2.2 Cross-price effect on x_2 when p_1 rises

Suppose the price of good 1 rises. Does the quantity demanded for good 2 rise / fall or cannot say? *Hint:* you can write the demand for good 2 as

$$x_2(p_1, p_2, m) = \frac{p_1}{p_1 + p_2} \cdot \frac{m}{p_2}.$$

Solution

Method 1: Differentiate the hinted expression

Holding $m, p_2 > 0$ fixed,

$$\frac{\partial x_2}{\partial p_1} = \frac{m}{p_2} \cdot \frac{(p_1 + p_2) - p_1}{(p_1 + p_2)^2} = \frac{m}{(p_1 + p_2)^2} > 0.$$

$$\boxed{\text{The quantity demanded of good 2 rises when the price of good 1 rises.}}$$

Method 2: Monotonicity of the fraction $p_1/(p_1 + p_2)$

If $p_1' > p_1$, then

$$\frac{p_1'}{p_1' + p_2} - \frac{p_1}{p_1 + p_2} = \frac{p_2(p_1' - p_1)}{(p_1' + p_2)(p_1 + p_2)} > 0,$$

so $x_2(p_1', p_2, m) > x_2(p_1, p_2, m)$.

$$\boxed{\text{The quantity demanded of good 2 rises when the price of good 1 rises.}}$$

2.3 Normal vs inferior goods

Are the goods normal, inferior, or neither?

Solution

Method 1: Income derivatives

$$x_1(p_1, p_2, m) = m \cdot \frac{p_2}{p_1(p_1 + p_2)}, \qquad x_2(p_1, p_2, m) = m \cdot \frac{p_1}{p_2(p_1 + p_2)}.$$

Thus

$$\frac{\partial x_1}{\partial m} = \frac{p_2}{p_1(p_1 + p_2)} > 0, \qquad \frac{\partial x_2}{\partial m} = \frac{p_1}{p_2(p_1 + p_2)} > 0,$$

so both goods are normal.

$$\boxed{\text{Both goods are normal.}}$$

Method 2: Homothetic scaling in income

For fixed prices, $x(p, m) = m\, a(p)$ for some $a(p) \in \mathbb{R}^2_{++}$. Hence $x(p, tm) = t\, x(p, m)$ for all $t > 0$, so both components increase with m.

$$\boxed{\text{Both goods are normal.}}$$

2.4 Slutsky substitution and income effects for good 1

The original price and expenditure is $p = (p_1, p_2)$ and m. Suppose the price of good 1 rises to $p_1' > p_1$ and so prices are $p' = (p_1', p_2)$. Calculate the Slutsky substitution and income effects. That is, calculate Δx_1^s and Δx_1^m. What can you say about the sign of these terms?

Solution

Method 1: Compute Slutsky effects and sign them

Marshallian demand for good 1 is

$$x_1(p_1, p_2, m) = \frac{mp_2}{p_1(p_1 + p_2)}.$$

Let $x^0 = x(p, m)$ and $m^s = p' \cdot x^0$. Define

$$\Delta x_1^s = x_1(p_1', p_2, m^s) - x_1(p_1, p_2, m), \qquad \Delta x_1^m = x_1(p_1', p_2, m) - x_1(p_1', p_2, m^s).$$

A direct calculation shows $x_1(p_1', p_2, m^s) < x_1(p_1, p_2, m)$, hence $\Delta x_1^s < 0$. Also $m^s = p' \cdot x^0 = m + (p_1' - p_1)x_1(p, m) > m$, so $m - m^s < 0$, implying $\Delta x_1^m < 0$.

$$\boxed{\Delta x_1^s < 0, \qquad \Delta x_1^m < 0 \text{ when } p_1' > p_1.}$$

Method 2: Sign logic + a numerical verification

An own-price increase implies a non-positive substitution effect; here it is strictly negative. Since good 1 is normal, the income effect is negative as well. As a check, if $p = (1, 1)$, $m = 2$, $p' = (2, 1)$, then $\Delta x_1^s = -\frac{1}{2}$ and $\Delta x_1^m = -\frac{1}{6}$, confirming both are negative.

$$\boxed{\Delta x_1^s < 0, \qquad \Delta x_1^m < 0.}$$

3 Perfect Complements

Consider the utility function

$$U(x_1, x_2) = \min\{x_1, x_2\}.$$

So, $U(2,3) = 2$, $U(6,1) = 1$, $U(100,3) = 3$ and so on.

3.1 Indifference curves

Draw some indifference curves for this utility function.

Solution

Method 1: Characterise indifference sets

Fix $\bar{u} \geq 0$. Then

$$U(x_1, x_2) = \bar{u} \iff \min\{x_1, x_2\} = \bar{u} \iff (x_1 \geq \bar{u},\ x_2 \geq \bar{u}) \text{ and at least one equals } \bar{u}.$$

So the indifference curve is the L-shaped boundary with kink at $(\bar{u}, \bar{u})$:

$$\{(\bar{u}, x_2) : x_2 \geq \bar{u}\} \ \cup \ \{(x_1, \bar{u}) : x_1 \geq \bar{u}\}.$$

$$\boxed{\text{Indifference curves are L-shaped with kink at } (\bar{u}, \bar{u}) \text{ on the } 45^\circ \text{ line.}}$$

Method 2: Fixed-proportions interpretation

Utility counts complete pairs; extra of one good beyond the minimum does not help, yielding L-shaped curves.

$$\boxed{\text{Perfect complements imply L-shaped indifference curves.}}$$

3.2 Marshallian demand

Find the demand function for this utility function. *Hint:* demand ought to satisfy $p{\cdot}x(p,m) = m$ and $x_1(p,m) = x_2(p,m)$.

Solution

Method 1: Use the hint $x_1 = x_2$ at optimum

Set $x_1 = x_2 = q$. Then $p_1 q + p_2 q = m \Rightarrow q = m/(p_1 + p_2)$, so

$$\boxed{x(p_1, p_2, m) = \left(\frac{m}{p_1 + p_2}, \ \frac{m}{p_1 + p_2}\right).}$$

Method 2: Maximise pairs subject to affordability

To achieve utility u at minimum cost choose (u, u), costing $(p_1 + p_2)u$. Maximise u subject to $(p_1 + p_2)u \leq m$, giving $u^* = m/(p_1 + p_2)$ and $x^* = (u^*, u^*)$.

$$\boxed{x(p_1, p_2, m) = \left(\frac{m}{p_1 + p_2}, \ \frac{m}{p_1 + p_2}\right).}$$

3.3 Normal vs inferior goods

Are the goods normal, inferior, or neither?

Solution

Method 1: Income derivatives

$$x_1(p,m) = x_2(p,m) = \frac{m}{p_1 + p_2} \ \Rightarrow \ \frac{\partial x_1}{\partial m} = \frac{\partial x_2}{\partial m} = \frac{1}{p_1 + p_2} > 0.$$

$$\boxed{\text{Both goods are normal.}}$$

Method 2: Linear scaling in income

Since $x(p, tm) = t\,x(p,m)$, both components increase with income.

$$\boxed{\text{Both goods are normal.}}$$

3.4 Slutsky substitution and income effects

The original price and expenditure is $p = (1, 1)$ and $m = 2$. Suppose the price of good 1 rises to 2 and so prices are $p' = (2, 1)$. Calculate the Slutsky substitution and income effects. That is, calculate Δx_1^s and Δx_1^m.

Solution

Method 1: Direct computation using the demand function

Initial:

$$x(p, m) = \left(\frac{2}{2}, \frac{2}{2}\right) = (1, 1) \;\Rightarrow\; x_1^0 = 1.$$

New Marshallian:

$$x(p', m) = \left(\frac{2}{3}, \frac{2}{3}\right) \;\Rightarrow\; x_1^1 = \frac{2}{3}.$$

Slutsky income and compensated bundle:

$$m^s = p' \cdot x(p, m) = (2, 1) \cdot (1, 1) = 3, \qquad x(p', m^s) = \left(\frac{3}{3}, \frac{3}{3}\right) = (1, 1) \Rightarrow x_1^c = 1.$$

Hence

$$\Delta x_1^s = x_1^c - x_1^0 = 0, \qquad \Delta x_1^m = x_1^1 - x_1^c = \frac{2}{3} - 1 = -\frac{1}{3}.$$

$$\boxed{\Delta x_1^s = 0, \qquad \Delta x_1^m = -\tfrac{1}{3}.}$$

Method 2: Zero substitution effect under Leontief

Perfect complements pin the compensated choice at the kink, so $\Delta x_1^s = 0$; all change comes from the income effect, $\Delta x_1^m = -1/3$.

$$\boxed{\Delta x_1^s = 0, \qquad \Delta x_1^m = -\tfrac{1}{3}.}$$